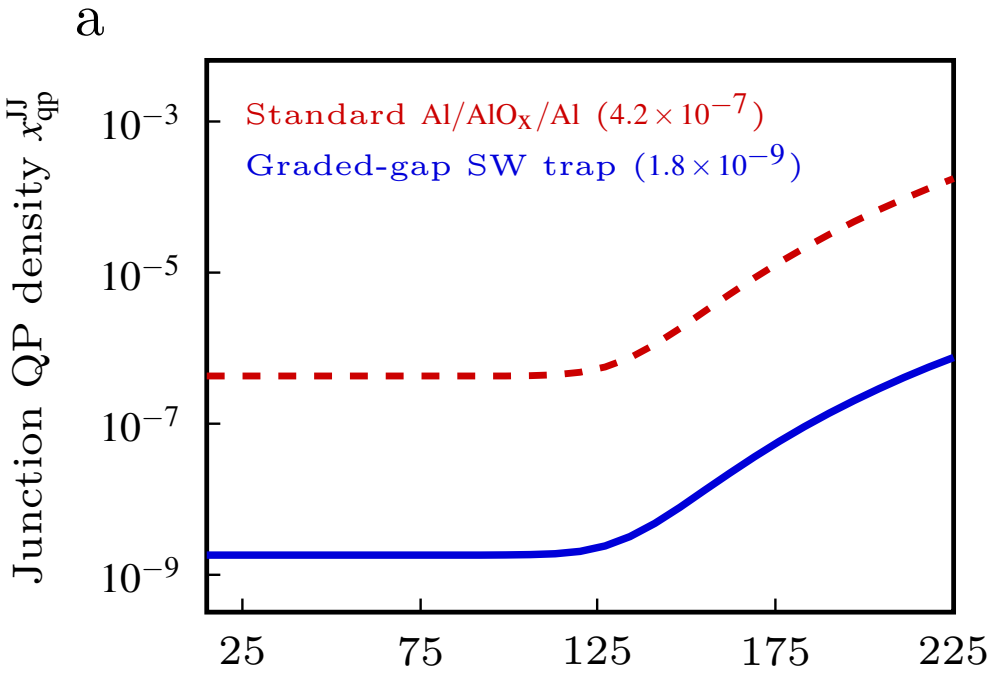
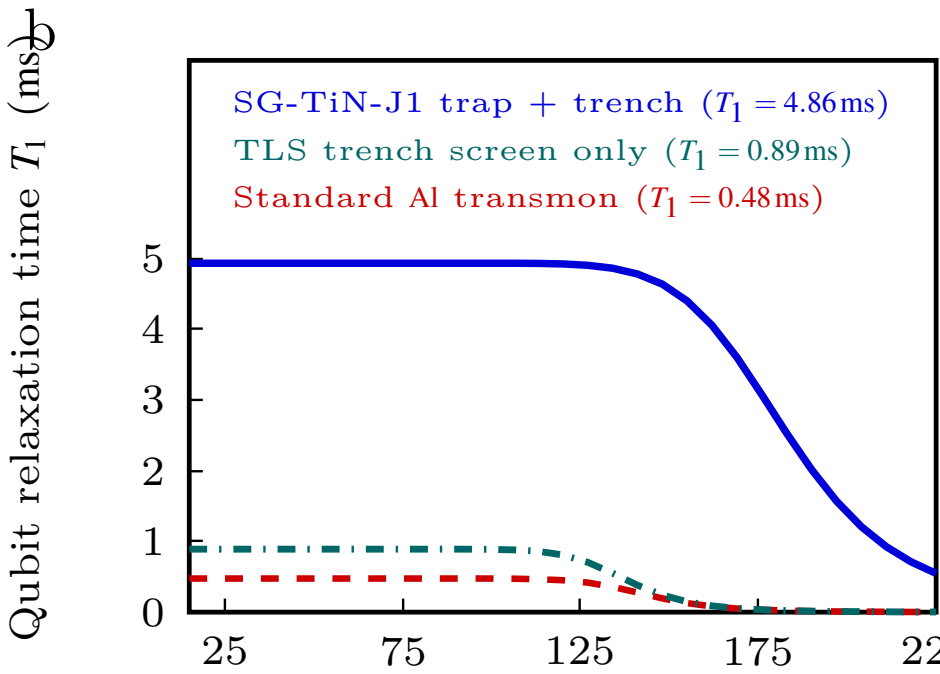

a
Standard Al/AlOx/Al (4.2 × 10−7)
Graded-gap SW trap (1.8 × 10−9)
Junction QP density x_{qp}^{JJ}
10−3
10−5
10−7
10−9
25
75
125
175
225
Mixing chamber temperature T_c (mK)


b
SG-TiN-J1 trap + trench (T_1 = 4.86 ms)
TLS trench screen only (T_1 = 0.89 ms)
Standard Al transmon (T_1 = 0.48 ms)
Qubit relaxation time T_1 (ms)
0
1
2
3
4
5
25
75
125
175
22
Mixing chamber temperature T_c (mK)